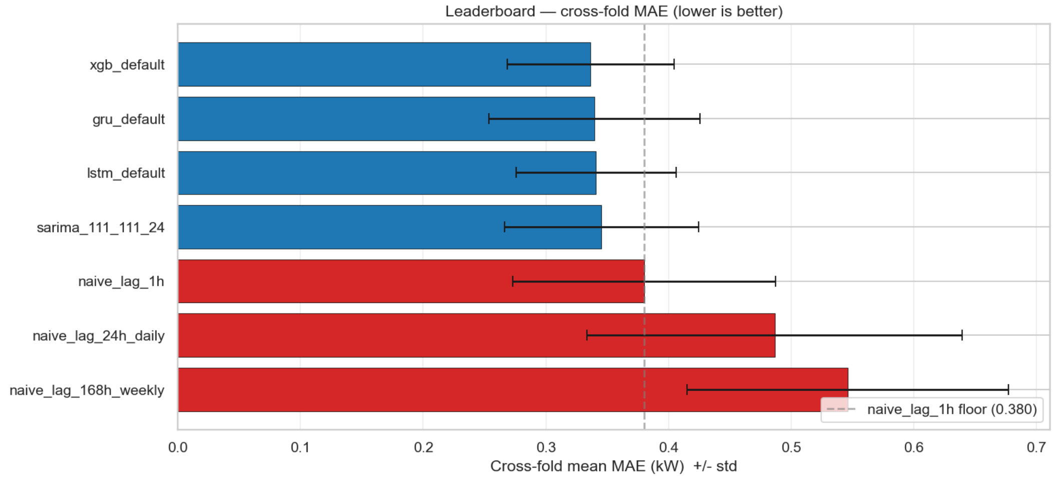
Forecasting Hourly Household Energy Consumption

LUISS Artificial Intelligence Techniques · Industry partner: Enel Global ICT · Thierry Ishimwe, Linda Carla Zorzoli, Mariavittoria Giurato, Cesar Dushimimana

MAE = 0.336 kW

Cross-fold mean, XGBoost (leakage-safe pipeline)

Four model families (SARIMA, XGBoost, LSTM, GRU) compared on a uniform rolling-origin protocol with five metrics. Same-timestamp UCI columns (*Voltage*, *Global_intensity*, sub-metering) are mechanically excluded from the feature matrix; short-gap imputation is past-only. The pipeline ships with two regression tests guarding both guarantees.

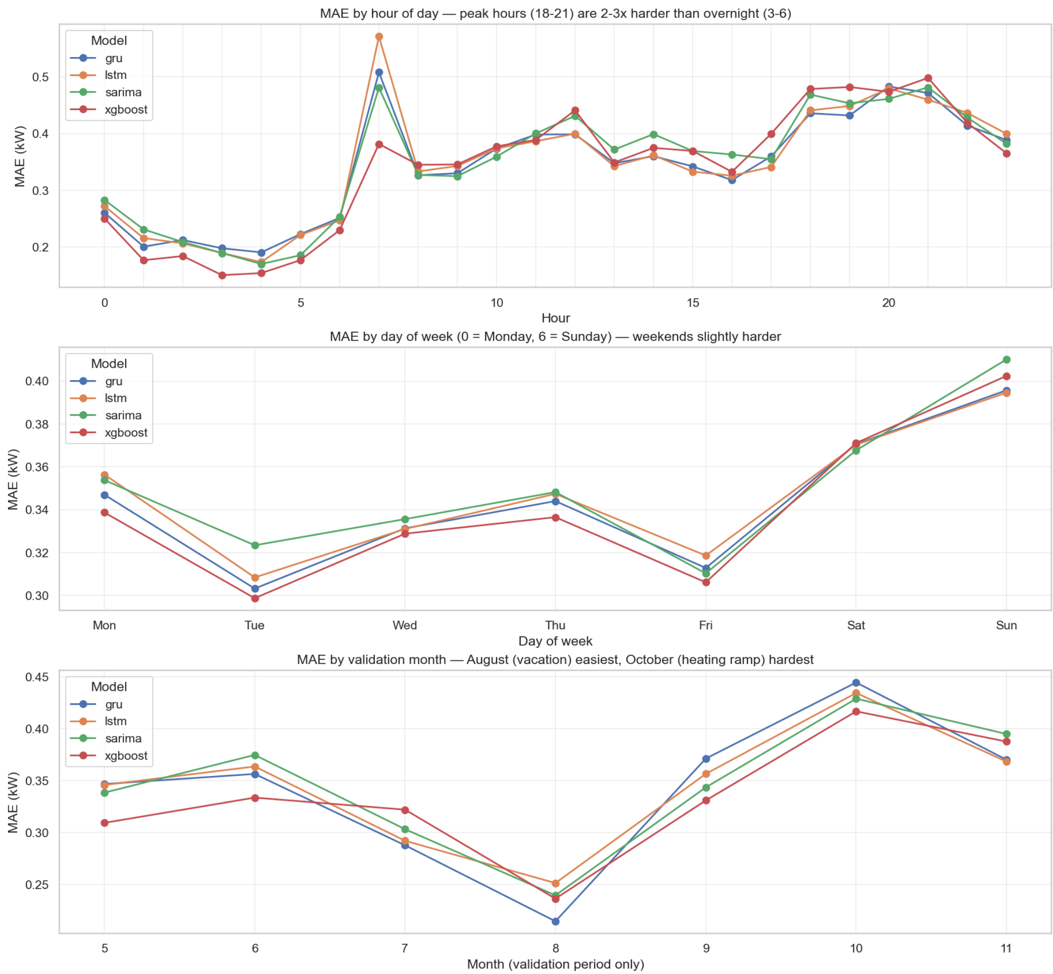


Headline findings

- **Top three are a statistical tie.** XGBoost 0.336, GRU 0.339, LSTM 0.341, SARIMA 0.345. Diebold–Mariano: XGBoost vs SARIMA ($p=0.022$) is the only significant pair at $\alpha=0.05$.
- **GRU competitive with LSTM at lower complexity.** Same MAE within noise; the simpler GRU cell captures sequential structure as well or better at this resolution.
- **Weather augmentation validates the future-work priority.** Adding 4 Paris–Montsouris weather columns to XGBoost: –0.77% MAE, concentrated on heating-season folds.
- **Leakage-safe by construction.** Registry blocks same-timestamp components of the target; past-only imputation; two regression tests guard the policy.

Where Models Win, Where They Lose, and What's Next

Per-segment performance, operational implications, and the priority queue



MAE by hour-of-day, day-of-week, and validation month. Peak hours and October's heating-ramp dominate the error budget.

Operational implications

- **Peak-hour MAE is ~2.5× trough-hour MAE.** XGBoost wins quiet hours; GRU wins peaks. For peak-cost-weighted operations, GRU may be the better deployment choice.
- **August easiest, October hardest.** August = *grandes vacances* (empty household, flat load). October = regulated *période de chauffe* (heating ramp, MAE ~2× higher).
- **Stacking does NOT break the 3% cluster.** Base-model errors are too correlated. The bottleneck is input information, not model variety.

Future work

- **Extend weather to SARIMA / LSTM / GRU.** The –0.77% gain on XGBoost is a lower bound. ~70 min DL retrain on the already-built augmented matrix.
- **24-step-ahead day-ahead forecasting.** Current results are 1-step-ahead at hourly resolution. Production at Enel uses 24-step horizons made once per day from a fixed cutoff.